Assignment 8

1. Show that if $\mathbb{Z}^r$ has an injective homomorphism to $\mathbb{Z}^s$, then $r \leq s$. Furthermore show that if such a homomorphism exists and $r = s$, then the image is a finite index subgroup of $\mathbb{Z}^s$.

2. Characterize those integers n for which all abelian groups of order n are cyclic.

3. List the elements of order 2 and 3 in $\mathbb{Z}_4 \oplus \mathbb{Z}_6$. Also find all the index 2 subgroups.

4. Prove that every finite abelian group is isomorphic to a direct product of cyclic groups of the form $\mathbb{Z}_{n_1} \oplus \mathbb{Z}_{n_2} \oplus \cdots \oplus \mathbb{Z}_{n_r}$, where $n_i | n_{i+1}$ for $i = 1, 2, \cdots, r-1$. (Note that this was proved for p-groups in the lectures)

5. Show that $\mathbb{Q}$ can be written as an increasing union of subgroups, each of which is a free group.